

Oracle Database 19c: SQL Tuning Workshop Training Course (3 days)

SQL Tuning in Oracle Database 19c is the process of improving SQL performance through analysis, statistics, and optimization techniques.

This instructor-led, live training (online or onsite) is aimed at intermediate-level Oracle professionals who wish to understand and apply SQL tuning techniques for Oracle Database 19c to improve query execution and database efficiency.

By the end of this training, participants will be able to:

- Interpret execution plans and identify performance bottlenecks.
- Apply Oracle tuning tools such as AWR, ADDM, and SQL Tuning Advisor.
- Use indexes, hints, and statistics to optimize queries.
- Diagnose and resolve inefficient SQL statements in real-world environments.

Format of the Course

- Interactive lecture and discussion.
- Hands-on use of Oracle tuning tools and SQL queries.
- Guided exercises focused on real tuning scenarios and performance analysis.

Course Outline

Introduction to SQL Tuning

- Performance tuning overview and goals
- Overview of Oracle Optimizer architecture
- Key tuning concepts: cost, cardinality, selectivity

Understanding Execution Plans

- Generating and interpreting execution plans
- EXPLAIN PLAN vs DBMS_XPLAN
- Common performance pitfalls in plans

Indexing Strategies

- Index types and their tuning implications
- Creating and analyzing indexes for performance
- Using invisible and function-based indexes

Oracle Tuning Tools

- Automatic Workload Repository (AWR)
- Automatic Database Diagnostic Monitor (ADDM)
- SQL Tuning Advisor and SQL Access Advisor

SQL Plan Management

- Plan baselines and capturing plans
- Managing plan evolutions
- Using SQL plan directives

Advanced SQL Tuning Techniques

- Bind peeking and adaptive cursor sharing
- Using hints and profiles to control execution
- Diagnosing and fixing complex queries

Hands-On Tuning Scenarios

- Analyzing real-world SQL problems
- Step-by-step tuning exercises
- Best practices and tuning checklists

Summary and Next Steps

Requirements

- An understanding of Oracle SQL and PL/SQL
- Experience working with Oracle Database as a developer or DBA
- Basic familiarity with execution plans and indexing concepts

Audience

- Oracle database developers
- Performance engineers
- Database administrators